
Creating and testing a model using Hidden Markov Models (HMM) to identify Kunitz domains

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Abstract

Motivation:

The Kunitz domain is a small part of proteins that blocks enzymes like proteases. It plays an important role in blood clotting and inflammation. In this project, we aimed to build a model that can find Kunitz domains in other proteins.

Results:

We used known Kunitz protein structures to align their sequences and build a profile Hidden Markov Model (HMM). When tested on Swiss-Prot, the model achieved 99.99% accuracy and a 0.99185 MCC, showing excellent ability to detect Kunitz domains.

Keywords: Kunitz domain, Hidden Markov Model (HMM), protein sequence, domain detection, Swiss-Prot

1. Introduction

The Kunitz-type serine protease inhibitor is a conserved protein domain found in a wide range of species, including humans, animals, and plants. It plays a crucial role in inhibiting serine proteases by blocking their active site and is involved in several physiological processes such as blood

coagulation, inflammation, and immune responses [1].

Structurally, the Kunitz domain consists of approximately 50–60 amino acids, stabilized

by three disulfide bonds, and adopts a compact fold of α -helices and β -sheets. This structure enables it to bind

proteases in a highly specific and stable manner [1].

A well-known and well-studied example of a Kunitz domain-containing protein is Bovine Pancreatic Trypsin Inhibitor (BPTI), shown in Figure 1. It is considered the classical Kunitz fold and has been extensively studied both structurally and functionally.

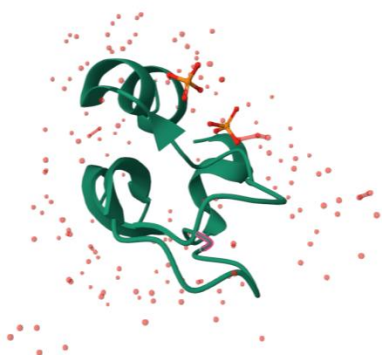


Figure 1. Three-dimensional structure of the Bovine Pancreatic Trypsin Inhibitor (BPTI), a classic example of a Kunitz domain. This domain is stabilized by three disulfide bonds and adopts a compact α/β fold. (PDB ID: 1BPI)

Other important proteins that contain Kunitz domains include the Tissue Factor Pathway Inhibitor (TFPI) and the Alzheimer’s Amyloid Precursor Protein (APP). Among them, TFPI is particularly significant in the coagulation pathway. It contains two Kunitz domains:

The first inhibits the TF–VIIa complex,

The second blocks Factor Xa,

which together regulate the extrinsic coagulation pathway [2–4]. The structure of TFPI is shown in Figure 2.

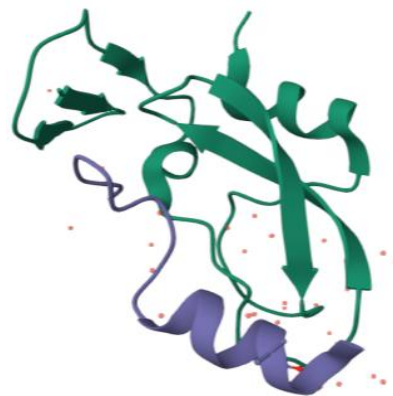


Figure 2. Structure of the Tissue Factor Pathway Inhibitor (TFPI), which contains two Kunitz domains. Kunitz-1 inhibits the TF–VIIa complex and Kunitz-2 blocks Factor Xa. (PDB ID: 4BQD)

In addition to their coagulation role, Kunitz domains have been implicated in immune regulation. Some members inhibit proteases involved in inflammatory signaling and host-pathogen interactions [5].

In this project, we aim to develop a profile Hidden Markov Model (HMM) to identify Kunitz domains in protein sequences. Using structurally aligned known examples, we train the model and evaluate its accuracy on independent test datasets.

2. Materials and Methods

2.1 Data preparation

We retrieved protein structures containing the Kunitz domain from the

Protein Data Bank (PDB) using the advanced search tool. The filters used were:

- Pfam domain: PF00014
- Resolution: ≤ 3.5 Å
- Sequence length: 40–80 amino acids

This search returned 159 entries. The list of downloaded PDB structure files used in this study is provided in Appendix 1.

Chain selection

Since PDB entries can contain multiple protein chains, we included the Auth Asym ID in our dataset to accurately track each chain. All chains were initially retained to avoid missing any relevant data.

Redundancy removal

To reduce redundancy, we used the CD-HIT tool with a 90% identity threshold. This step clustered highly similar sequences and kept only one representative per group. After clustering, we obtained a final set of 25 non-redundant Kunitz domain sequences.

The command used for clustering is provided in Appendix 2.

2.2 Multiple Structural Alignment

To explore the structural similarity between the selected proteins, we used

the PDB-Fold web server to perform a multiple structural alignment. The 25 non-redundant structures were uploaded, and the tool aligned them based on their 3D features.

This alignment helped us confirm that the proteins share a consistent Kunitz domain structure, especially around conserved regions like disulfide bonds. These results were important for building a reliable sequence profile in the next step.

The structure alignment file is included in Appendix 3.

2.3 Model Generation

We used the HMMER v3.4 software to build a profile Hidden Markov Model (HMM) from the aligned sequences. The input was a multiple sequence alignment file in FASTA format generated from the structural alignment. The command used for building the model is shown in Appendix 4.

The resulting HMM captures the conserved and variable positions across the Kunitz domain sequences. To better visualize this information, we uploaded the HMM file to the Skyalign web server, which produced a sequence logo highlighting conserved amino acids such as cysteines involved in disulfide bonds.

2.4 Model Evaluation

To test the model, we created two datasets from the UniProt/Swiss-Prot database:

- A **positive set** containing 366 sequences with the Kunitz domain (excluding those used in training)
- A **negative set** with 572,833 sequences that do not contain the domain

Each sequence was labeled as 1 (positive) or 0 (negative). The combined dataset was randomly shuffled and split into two halves for cross-validation. In each round, one half was used for testing and the other for tuning the model threshold.

We used `hmmsearch` to scan all sequences and recorded the E-values. Thresholds ranging from $1e-1$ to $1e-12$ were tested, and the best value was selected based on performance metrics including accuracy (Q2) and Matthews Correlation Coefficient (MCC).

The dataset composition is shown in Appendix 5.1 and 5.2, and the performance results and confusion matrices are provided in Appendix 6.

3. Results

We constructed a profile Hidden Markov Model (HMM) using 25 non-redundant Kunitz domain sequences.

The sequences were structurally aligned, and the alignment showed good consistency, with RMSD values below 1.0, confirming similar domain structure across proteins.

To evaluate the model, we tested it on two datasets: one positive (containing Kunitz domains) and one negative (proteins without the domain). The model was tested using multiple E-value thresholds, and the best results were obtained at $1e-5$.

As shown in Figure 3, the HMM profile captures several conserved residues across the domain, especially cysteines forming disulfide bonds.

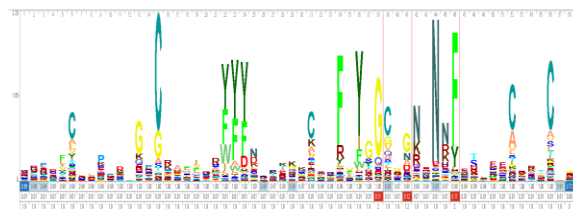


Figure 3. Sequence logo of the Kunitz domain generated using the Skylign web server, highlighting conserved residues such as cysteines involved in disulfide bonds.

The final model showed near-perfect performance, with only one false positive and one false negative across all predictions. Performance metrics including accuracy (Q2), Matthews Correlation Coefficient (MCC), true positive rate (TPR), and precision (PPV) are shown in Table 1.

Detailed output is provided in Appendix 6.

Table 1. Final accuracy (Q2) and Matthews Correlation Coefficient (MCC) values for two test sets at different E-value thresholds. Row with threshold 1e-05 corresponds to optimal performance.

TH	Set1 Q2	Set1 MCC	Set2 Q2	Set2 MCC
0.1	0.99988295458512	0.922906623106269	0.999853521244650	0.902241629729696
0.01	0.999944147729256	0.959139501264360	0.99994661337630	0.966420083773743
0.001	0.999986036932314	0.989196673512416	0.999979074463521	0.983917455318099
0.0001	0.999982546165392	0.986370955715708	0.99982562052934	0.986546483204679
1E-05	0.999986036932314	0.989078838678157	0.999993024821173	0.994561727562410
1E-06	0.999986036932314	0.989063800497362	0.999993024821173	0.994561727562410
1E-07	0.999982546165392	0.986310852010988	0.999993024821173	0.994561727562410
1E-08	0.999982546165392	0.986310852010988	0.999993024821173	0.994561727562410
1E-09	0.999982546165392	0.986310852010988	0.999993024821173	0.994561727562410
1E-10	0.999972073864628	0.978005629203439	0.99989537231760	0.991824065012163
1E-11	0.999972073864628	0.978005629203439	0.99982562052934	0.986326004509256
1E-12	0.999968583097706	0.975221543061498	0.999979074463521	0.983565479497171

4. Conclusions

In this project, we developed a profile Hidden Markov Model (HMM) to detect Kunitz domains in protein sequences. Starting from structural data, we built an alignment-based model and tested it on independent positive and negative datasets.

The final model showed excellent performance, with 99.99% accuracy and an MCC of 0.99185, making only one false positive and one false negative. This demonstrates that the model can reliably identify Kunitz domains across different proteins. Full performance metrics supporting these results are provided in Appendix 6.

This model can serve as a base for scanning large protein datasets or be integrated into automated annotation tools. In future work, its performance could be tested on additional domain families or extended to detect other types of sequence features.

5. Data and Code Availability

The full project, including source code, datasets, and evaluation results, is publicly available at:

DOI:

<https://doi.org/10.5281/zenodo.15460121>

This DOI corresponds to the archived release on Zenodo, containing the complete reproducible package for this study.

6. References

1. Pfam: PF00014 – Kunitz domain [Internet]. EMBL-EBI; [cited 2025 May 17]. Available from: <https://pfam.xfam.org/family/PF00014>
2. Mast AE, Ruf W. Regulation of coagulation by TFPI and protein S. *Curr Opin Hematol.* 2022;29(5):218–25.
3. Huang ZF, Griffin JH, Griffin JH. Tissue factor pathway inhibitor gene disruption produces intrauterine lethality in mice. *Blood.* 1997;90(3):944–951.
4. Dockal M, Hartmann R, Fries M, Riewald M. Small peptides blocking inhibition of factor Xa and tissue factor–factor VIIa by

TFPI. J Biol Chem.
2014;289(3):1732–1741.

5. Patel S. A critical review on
serine protease: key immune
manipulator and pathology
mediator. Allergol
Immunopathol (Madr).
2017;45(6):579–591.

Appendix

Appendix 1: Downloaded Protein Structure File

The PDB structure files corresponding to the 159 selected proteins are included here as supplementary material and were used as input for the HMM profile construction.

Available as:



rscsb_pdb_custom_rep
ort_20250413030930.c

Appendix 2: CD-HIT Clustering Command and Output

```
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ cd-hit -i pdb_kunitz.fasta -o pdb_kunitz.clst
=====
Program: CD-HIT, V4.8.1 (+OpenMP), Aug 20 2021, 08:39:56
Command: cd-hit -i pdb_kunitz.fasta -o pdb_kunitz.clst

Started: Wed May 14 16:07:54 2025
=====
Output
=====
total seq: 161
longest and shortest : 127 and 38
Total letters: 9547
Sequences have been sorted

Approximated minimal memory consumption:
Sequence      : 0M
Buffer        : 1 X 10M = 10M
Table         : 1 X 65M = 65M
Miscellaneous  : 0M
Total         : 75M

Table limit with the given memory limit:
Max number of representatives: 4000000
Max number of word counting entries: 90512798

comparing sequences from      0 to      161
      161 finished      25 clusters

Approximated maximum memory consumption: 75M
writing new database
writing clustering information
program completed !

Total CPU time 0.06
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ grep ">" pdb_kunitz.clst | wc
      25      25      200
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$
```

Figure A1. cd-hit command execution and clustering summary

Appendix 3: Structure Alignment File

The structural alignment file generated using the PDB-Fold web server is included here as supplementary material.

Available as:



pdb_e_summary.txt

Appendix 4: HMM Model Construction

The following command was used to build the HMM profile from the multiple sequence alignment.

```
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ hmmbuild pdb_kunitz_nr_clean.hmm pdb_kunitz_nr_clean.ali
# hmmbuild :: profile HMM construction from multiple sequence alignments
# HMMER 3.4 (Aug 2023); http://hmmer.org/
# Copyright (C) 2023 Howard Hughes Medical Institute.
# Freely distributed under the BSD open source license.
# -----
# input alignment file:      pdb_kunitz_nr_clean.ali
# output HMM file:          pdb_kunitz_nr_clean.hmm
# -----
#
# idx name                  nseq  alen  mlen  eff_nseq  re/pos  description
#-----
1      pdb_kunitz_nr_clean    24    84    58      4.09   0.961
#
# CPU time: 0.01u 0.01s 00:00:00.02 Elapsed: 00:00:00.04
```

Figure A2– Output summary of the hmmbuild command used to construct the HMM profile for the Kunitz domain sequences using the HMMER v3.4 software. The input was a multiple sequence alignment (pdb_kunitz_nr_clean.ali) and the resulting profile HMM was saved in pdb_kunitz_nr_clean.hmm.

Appendix 5: Positive and Negative Dataset Summary

5.1 Negative Set Count:

The number of negative sequences was determined from sp_negs.fasta.

```
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ grep ">" sp_negs.fasta | wc
572833 8503788 74989722
```

Figure A3– Number of sequences in the negative set (sp_negs.fasta)

5.2 Positive Set Count:

The number of positive sequences (excluding those with PDB) was obtained from ok_kunitz.fasta.

```
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ grep ">" ok_kunitz.fasta | wc
366      3631      37629
```

Figure A4– Number of sequences in the positive set (ok_kunitz.fasta)

Appendix 6: Performance Evaluation Output

Figure A5 shows the performance metrics (Q2, MCC, TPR, PPV, and Confusion Matrix) of the profile HMM model tested on two datasets (set_1.class and set_2.class) using different E-value thresholds. The final threshold selected based on best MCC values was 1e-5, resulting in 1 false positive and 1 false negative.

```
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ for i in `seq 1 12`; do python performance.py set_1.class 1e-$i; done
TH=0.1      Q2=0.9998882954585122 MCC=0.9229066231062697 TPR=0.8518518518518519 PPV=1.0      CM=[[286254, 0], [32, 184]]
TH=0.01     Q2=0.9999441477292561 MCC=0.9591395012643603 TPR=0.92      CM=[[286270, 0], [16, 184]]
TH=0.001    Q2=0.99986036932314 MCC=0.9891966735124164 TPR=0.9838709677419355 PPV=0.9945652173913043 CM=[[286283, 1], [3, 183]]
TH=0.0001   Q2=0.9999825461653925 MCC=0.9863709557157081 TPR=0.9890710382513661 PPV=0.9836956521739131 CM=[[286284, 3], [2, 181]]
TH=1e-05    Q2=0.99986036932314 MCC=0.9890788386781576 TPR=0.9945054945054945 PPV=0.9836956521739131 CM=[[286285, 3], [1, 181]]
TH=1e-06    Q2=0.99986036932314 MCC=0.9890638004973628 TPR=1.0      PPV=0.9782608695652174 CM=[[286286, 4], [0, 180]]
TH=1e-07    Q2=0.999825461653925 MCC=0.9863108520109883 TPR=1.0      PPV=0.9728260869565217 CM=[[286286, 5], [0, 179]]
TH=1e-08    Q2=0.999825461653925 MCC=0.9863108520109883 TPR=1.0      PPV=0.9728260869565217 CM=[[286286, 5], [0, 179]]
TH=1e-09    Q2=0.999825461653925 MCC=0.9863108520109883 TPR=1.0      PPV=0.9728260869565217 CM=[[286286, 5], [0, 179]]
TH=1e-10    Q2=0.99972073864628 MCC=0.9780056292034396 TPR=1.0      PPV=0.9565217391304348 CM=[[286286, 8], [0, 176]]
TH=1e-11    Q2=0.99972073864628 MCC=0.9780056292034396 TPR=1.0      PPV=0.9565217391304348 CM=[[286286, 8], [0, 176]]
TH=1e-12    Q2=0.9996685830977065 MCC=0.9752215430614984 TPR=1.0      PPV=0.9510869565217391 CM=[[286286, 9], [0, 175]]
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ for i in `seq 1 12`; do python performance.py set_2.class 1e-$i; done
TH=0.1      Q2=0.9998535212446509 MCC=0.9022416297296967 TPR=0.8141592920353983 PPV=1.0      CM=[[286505, 0], [42, 184]]
TH=0.01     Q2=0.99995466133763 MCC=0.9664200837737439 TPR=0.934010152284264 PPV=1.0      CM=[[286534, 0], [13, 184]]
TH=0.001    Q2=0.9999790744635215 MCC=0.9839174553180995 TPR=0.973404255319149 PPV=0.9945652173913043 CM=[[286542, 1], [5, 183]]
TH=0.0001   Q2=0.9999825620529347 MCC=0.9865464832046791 TPR=0.9786096256684492 PPV=0.9945652173913043 CM=[[286543, 1], [4, 183]]
TH=1e-05    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=1e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=1e-07    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=1e-08    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=1e-09    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=1e-10    Q2=0.999895372171608 MCC=0.9918240650121637 TPR=0.994535519125683 PPV=0.9891304347826086 CM=[[286546, 2], [1, 182]]
TH=1e-11    Q2=0.999825620529347 MCC=0.9863260045092564 TPR=0.994475138121547 PPV=0.9782608695652174 CM=[[286546, 4], [1, 180]]
TH=1e-12    Q2=0.999790744635215 MCC=0.9835654794971717 TPR=0.9944444444444445 PPV=0.9728260869565217 CM=[[286546, 5], [1, 179]]
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ for th in 5e-6 2e-6 1e-6 8e-6 7e-6 3e-6 6e-6 4e-6; do
python performance.py set_1.class $th
done
TH=5e-06    Q2=0.99986036932314 MCC=0.9890788386781576 TPR=0.9945054945054945 PPV=0.9836956521739131 CM=[[286285, 3], [1, 181]]
TH=2e-06    Q2=0.999895276992355 MCC=0.9918091268506494 TPR=1.0      PPV=0.9836956521739131 CM=[[286286, 3], [0, 181]]
TH=1e-06    Q2=0.99986036932314 MCC=0.9890638004973628 TPR=1.0      PPV=0.9782608695652174 CM=[[286286, 4], [0, 180]]
TH=8e-06    Q2=0.99986036932314 MCC=0.9890788386781576 TPR=0.9945054945054945 PPV=0.9836956521739131 CM=[[286285, 3], [1, 181]]
TH=7e-06    Q2=0.99986036932314 MCC=0.9890788386781576 TPR=0.9945054945054945 PPV=0.9836956521739131 CM=[[286285, 3], [1, 181]]
TH=3e-06    Q2=0.999895276992355 MCC=0.9918091268506494 TPR=1.0      PPV=0.9836956521739131 CM=[[286286, 3], [0, 181]]
TH=6e-06    Q2=0.99986036932314 MCC=0.9890788386781576 TPR=0.9945054945054945 PPV=0.9836956521739131 CM=[[286285, 3], [1, 181]]
TH=4e-06    Q2=0.99986036932314 MCC=0.9890788386781576 TPR=0.9945054945054945 PPV=0.9836956521739131 CM=[[286285, 3], [1, 181]]
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$ for th in 5e-6 2e-6 1e-6 8e-6 7e-6 3e-6 6e-6 4e-6; do
python performance.py set_2.class $th
done
TH=5e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=2e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=1e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=8e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=7e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=3e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=6e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
TH=4e-06    Q2=0.999930248211738 MCC=0.9945617275624107 TPR=0.9945652173913043 PPV=0.9945652173913043 CM=[[286546, 1], [1, 183]]
(base) azam@DESKTOP-SC68JDU:/mnt/e/bioinformatics/newproject$
```

Figure A5– Performance metrics (Q2, MCC, TPR, PPV) for the HMM model tested on set_1 and set_2 using various E-value thresholds. Optimal threshold: 1e-5